

Lecture 24 -- Reasoning Patterns

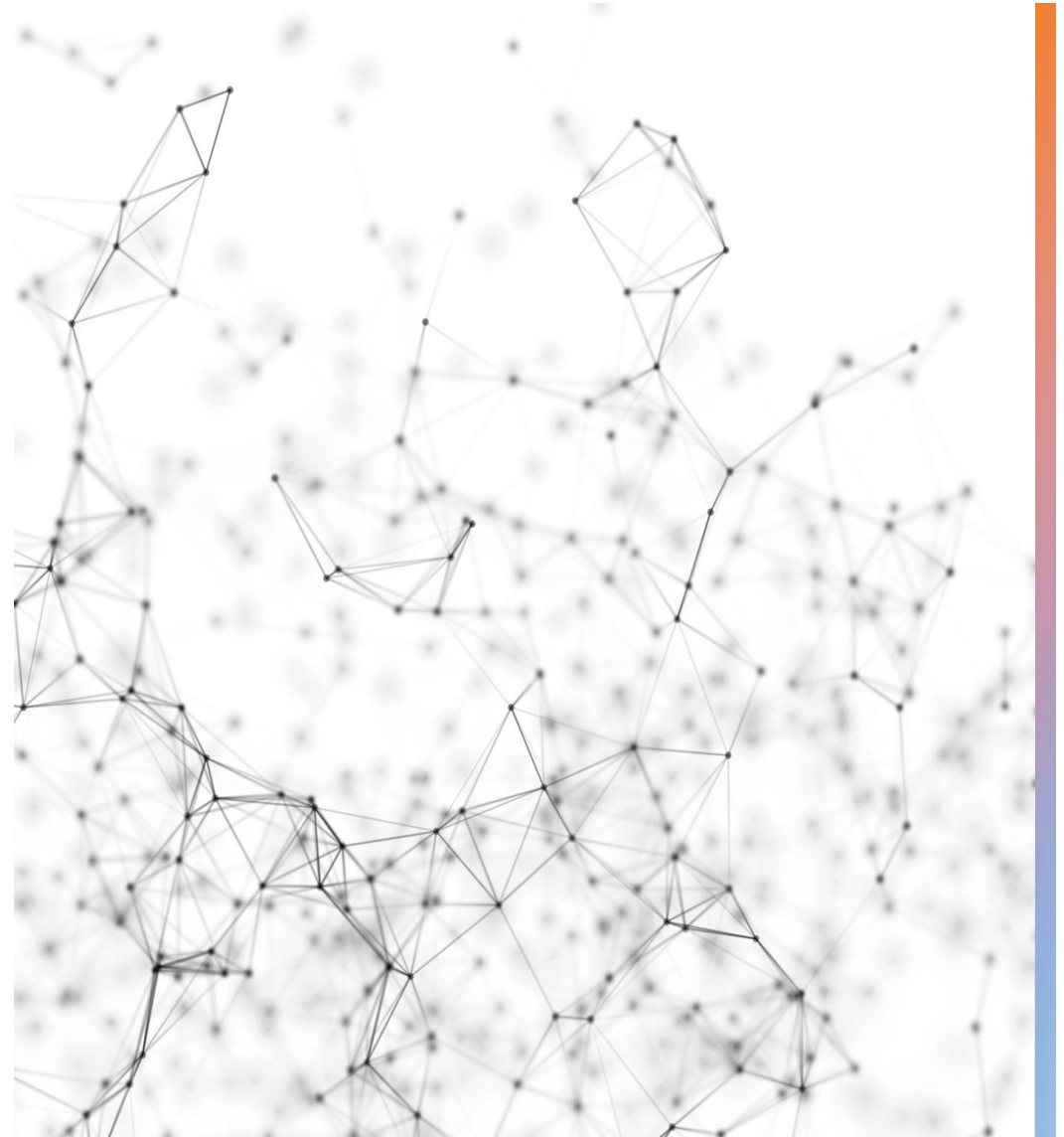
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(Attendance Code: **282462**)

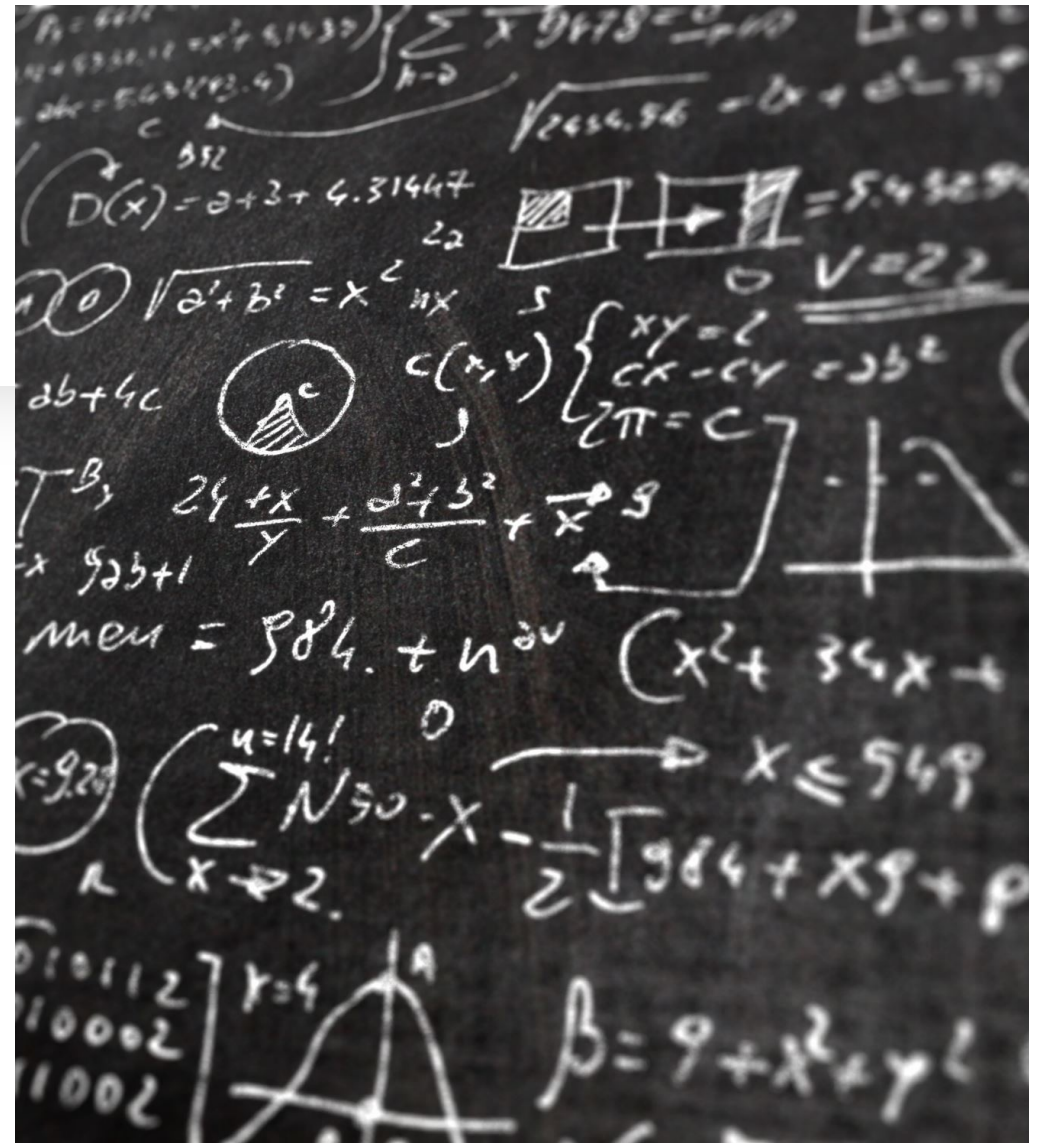
Up to now,

- Traditional Machine Learning Algorithms
- Adversarial Attack and Defence
- Deep learning
- Probabilistic Graphical Models
 - Introduction
 - Data $\Rightarrow$ Distribution $\Leftrightarrow$ Graph



Topics

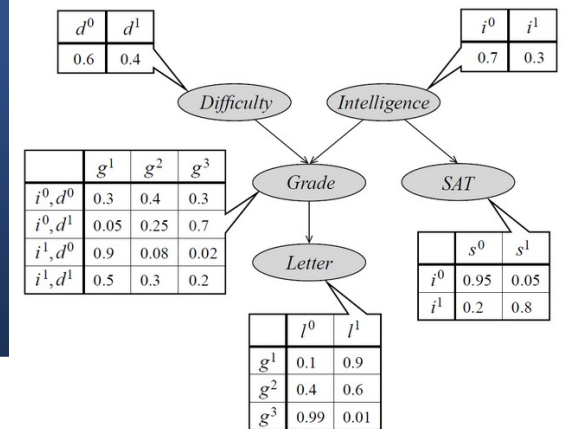
- Reasoning Patterns (i.e., do reasoning on a given graph)
 - Causal Reasoning
 - Evidential Reasoning
 - Intercausal reasoning



Recap: Local Independencies in a BN

- A BN G is a directed acyclic graph whose nodes represent random variables $X_1, \dots, X_n$.
- Let $Pa(X_i)$ denote parents of X_i in G
- Let $Non-Desc(X_i)$ denote variables in G that are not descendants of X_i
- Then G encodes the following set of *conditional independence* assumptions denoted $I(G)$
 - For each X_i : $(X_i \perp Non-Desc(X_i) \mid Pa(X_i))$
- Also known as *Local Markov Independencies*

Recap: Local Independencies



- Graph G with CPDs is equivalent to a set of independence assertions

$$P(D, I, G, S, L) = P(D)P(I)P(G|D, I)P(S|I)P(L|G)$$

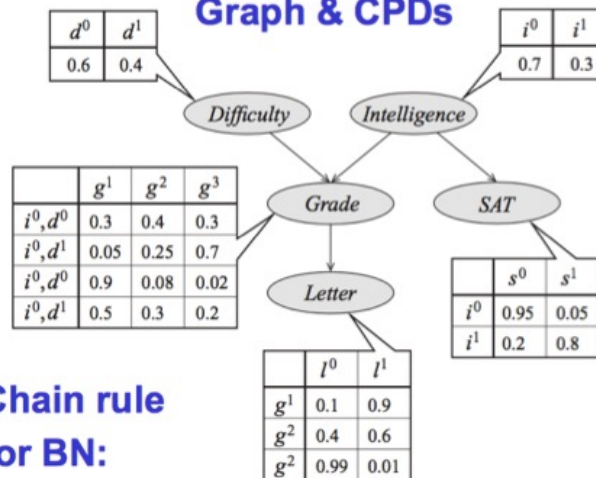
- Local Conditional Independence Assertions (starting from leaf nodes):

$I_{local}(G) = \{$
 $(L \perp I, D, S | G),$ L is conditionally independent of all other nodes given parent G
 $(S \perp D, G, L | I),$ S is conditionally independent of all other nodes given parent I
 $(G \perp S | D, I),$ Even given parents, G is NOT independent of descendant L
 $(I \perp D | \emptyset),$ Nodes with no parents are marginally independent
 $(D \perp I, S | \emptyset)\}$ D is independent of non-descendants I and S

- Parents of a variable shield it from probabilistic influence
 - Once value of parents known, no influence of ancestors
- Information about descendants can change beliefs about a node

Recap: Evaluating a Joint Probability

Graph & CPDs



$Val(I) = \{i^0 = \text{low intelligence}, i^1 = \text{high intelligence}\}$

$Val(D) = \{d^0 = \text{easy}, d^1 = \text{hard}\}$

$Val(G) = \{g^1 = A, g^2 = B, g^3 = C\}$

$Val(S) = \{s^0 = \text{low}, s^1 = \text{high}\}$

$Val(L) = \{l^0 = \text{weak}, l^1 = \text{strong}\}$

Chain rule for BN:

$$P(D, I, G, S, L) = P(D)P(I)P(G|D, I)P(S|I)P(L|G)$$

$$P(i^1, d^0, g^2, s^1, l^0) = P(i^1)P(d^0)P(g^2|i^1, d^0)P(s^1|i^1)P(l^0|g^2)$$

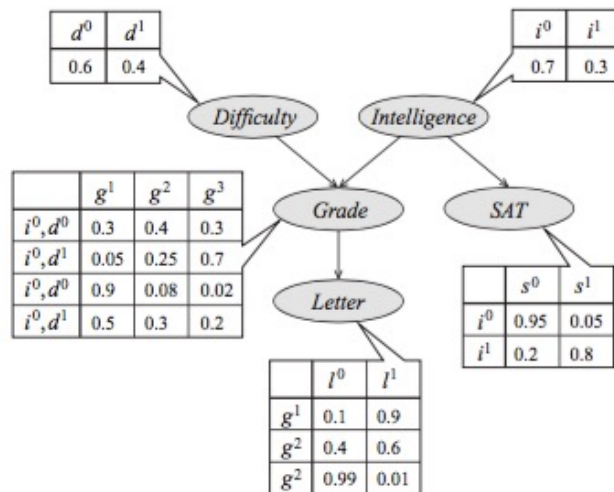
$$= 0.3 \cdot 0.6 \cdot 0.08 \cdot 0.8 \cdot 0.4 = 0.004608$$



$P(\text{high intelligence, easy course, grade=B, high SAT, weak letter}) = \text{very low}$

Reasoning Patterns

- Reasoning about a student *George* using the model



Causal Reasoning

- George* is interested in knowing as to how likely he is to get a strong *Letter* (based on *Intelligence*, *Difficulty*)?

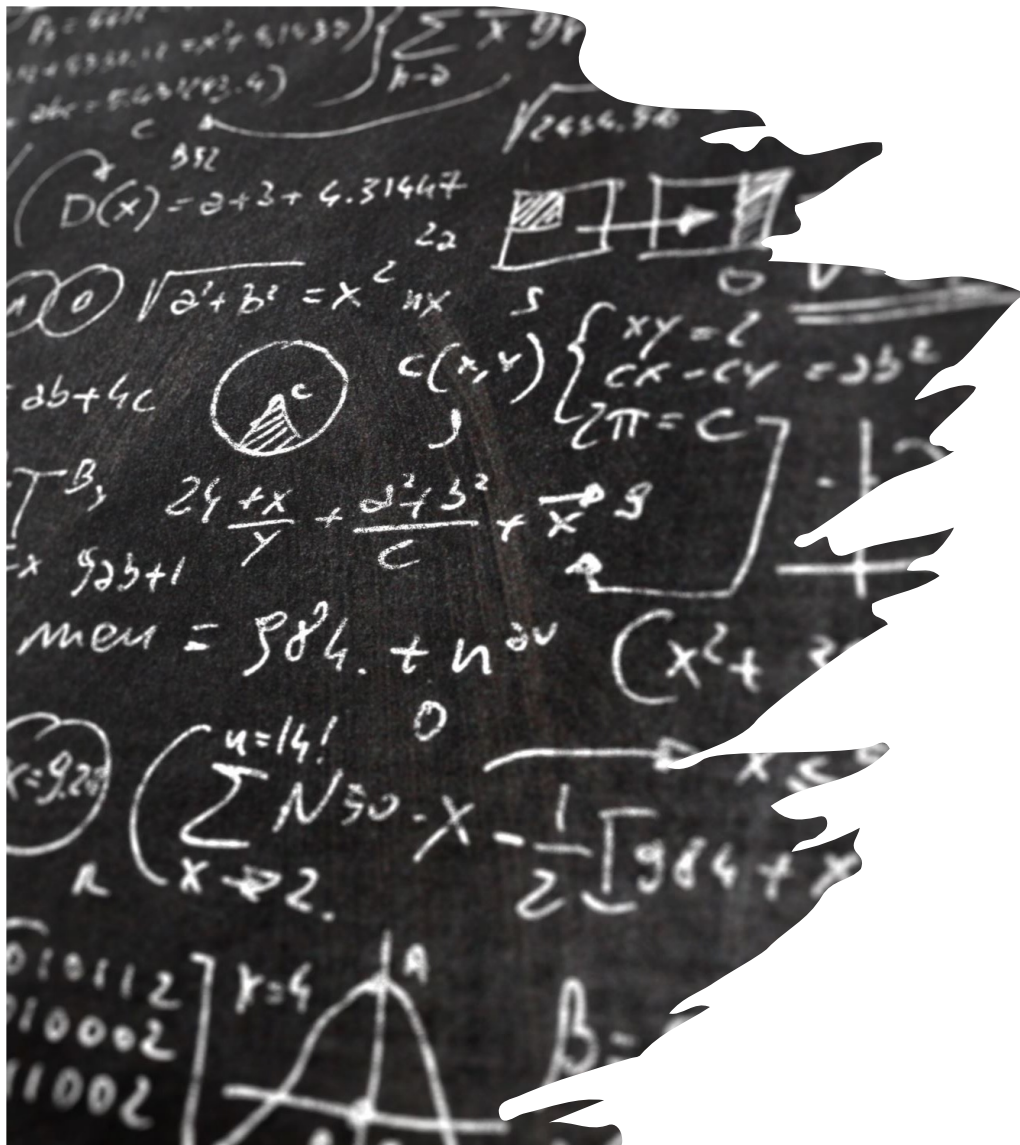
Evidential Reasoning

- Recruiter* is interested in knowing whether *George* is *Intelligent* (based on *Letter*, *SAT*)

George

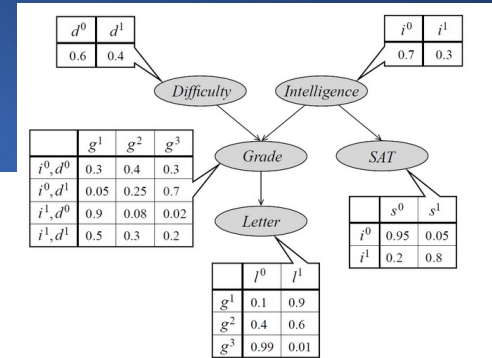


Recruiter



Causal Reasoning

Causal Reasoning



- How likely *George* will get a strong *Letter* (No evidence)?

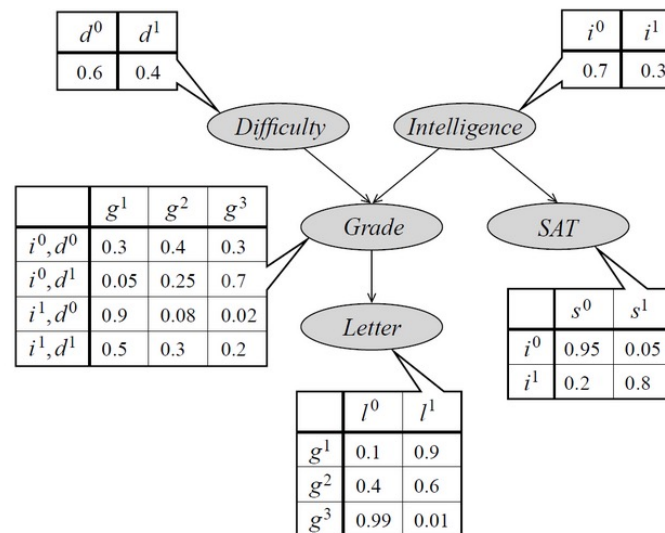
$$\begin{aligned}
 P(l^1) &= \sum_{D,I,G,S} P(D, I, G, S, L = l^1) = \sum_{D,I,G,S} P(D)P(I)P(G|D, I)P(S|I)P(l^1|G) \\
 &= \begin{aligned}
 &P(D = d^0)P(I = i^0)P(g = g1|D = d^0, I = i^0)P(S = s^0|I = i^0)P(L = l^1|g = g1) \\
 &+ P(D = d^0)P(I = i^0)P(g = g2|D = d^0, I = i^0)P(S = s^0|I = i^0)P(L = l^1|g = g2) \\
 &+ P(D = d^0)P(I = i^0)P(g = g3|D = d^0, I = i^0)P(S = s^0|I = i^0)P(L = l^1|g = g3) \\
 &+ P(D = d^0)P(I = i^0)P(g = g1|D = d^0, I = i^0)P(S = s^1|I = i^0)P(L = l^1|g = g1) \\
 &+ P(D = d^0)P(I = i^0)P(g = g2|D = d^0, I = i^0)P(S = s^1|I = i^0)P(L = l^1|g = g2) \\
 &+ P(D = d^0)P(I = i^0)P(g = g3|D = d^0, I = i^0)P(S = s^1|I = i^0)P(L = l^1|g = g3) \\
 &\dots
 \end{aligned}
 \end{aligned}$$

Causal Reasoning

- How likely *George* will get a strong *Letter* (No evidence)?

$$P(l^1) = \sum_{D,I,G,S} P(D, I, G, S, L = l^1) = \sum_{D,I,G,S} P(D)P(I)P(G|D, I)P(S|I)P(l^1|G)$$

- $P(l^1)=0.502$
- Obtained by summing-out other variables in joint distribution



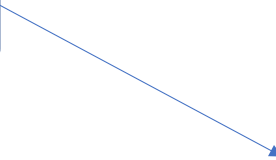
Causal Reasoning

- Knowing *George* is not so *Intelligent* (i^0)

$$P(l^1|i^0) = \frac{P(l^1, i^0)}{P(i^0)} = \frac{\sum_{D,I,G} P(D)P(i^0)P(G|D, i^0)P(S|i^0)P(l^1|G)}{\sum_{D,G,S,L} P(D)P(i^0)P(G|D, i^0)P(S|i^0)P(L|G)}$$

- $P(l^1|i^0)=0.389$

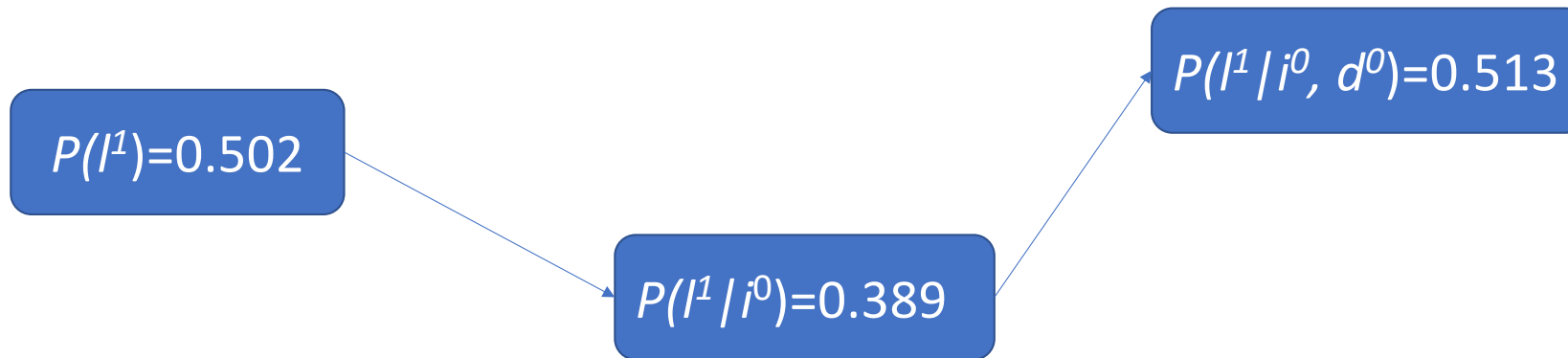
$$P(I^1)=0.502$$


$$P(I^1 | i^0)=0.389$$

After knowing that the student is not as intelligent, we understand that the probability of getting a strong recommendation letter is lowered.

Causal Reasoning

- Knowing COMP219 is not *Difficult* (d^0)
- $P(I^1 / i^0, d^0) = 0.513$ (Exercise!)

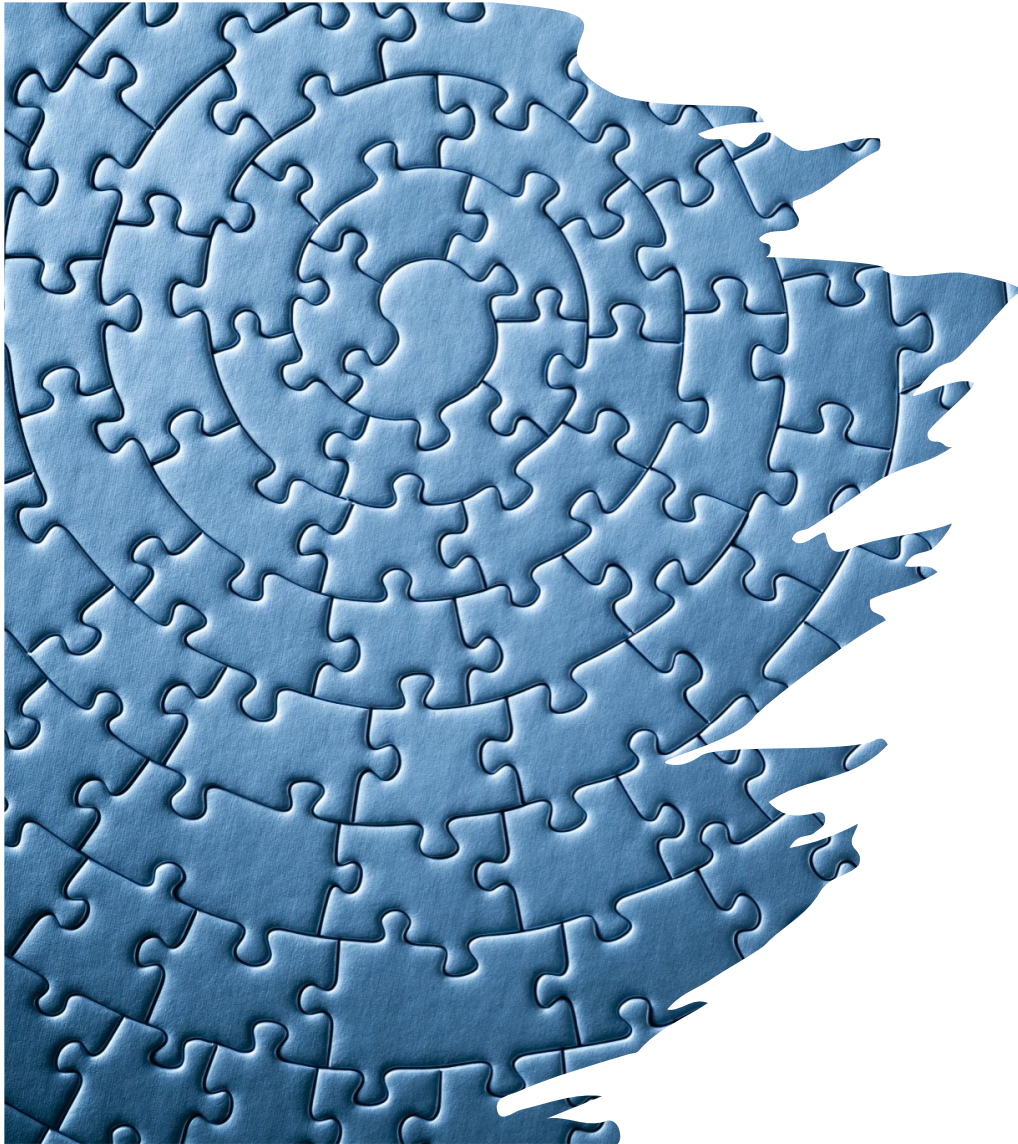


After knowing that the student is not as intelligent, we understand that the probability of getting a strong recommendation letter is lower.

After further knowing that the difficulty is low, the probability of getting a strong letter is higher.

Causal Reasoning

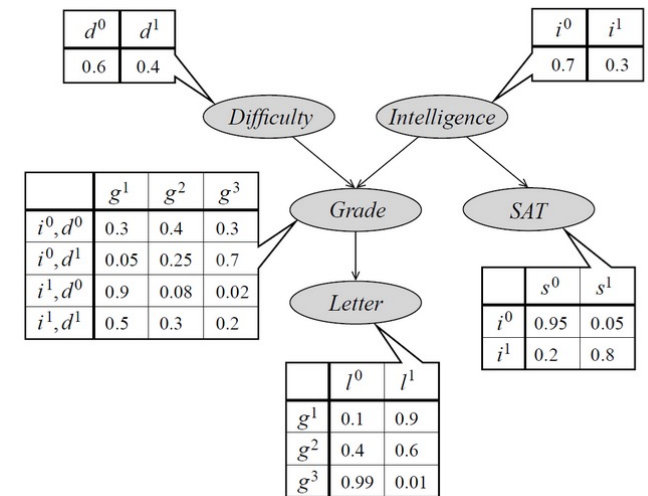
- Observe how probabilities change as more evidence is obtained
- ***Causal Reasoning:***
Predicting downstream effects of factors such as *Intelligence*

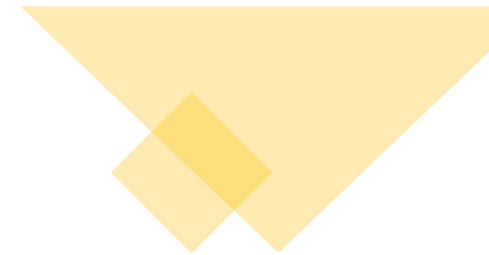
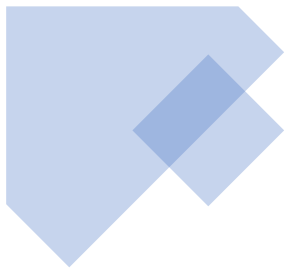


Evidential Reasoning

Evidential Reasoning

- Recruiter wants to hire *Intelligent* student
- A priori *George* is 30% likely to be *Intelligent*
 $P(i^1)=0.3$
- Finds that *George* received *Grade C* (g^3) in *COMP219*
 $P(i^1 / g^3)=0.079$



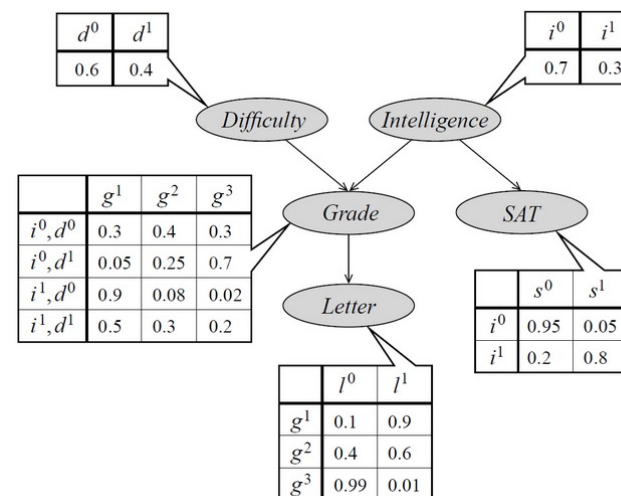


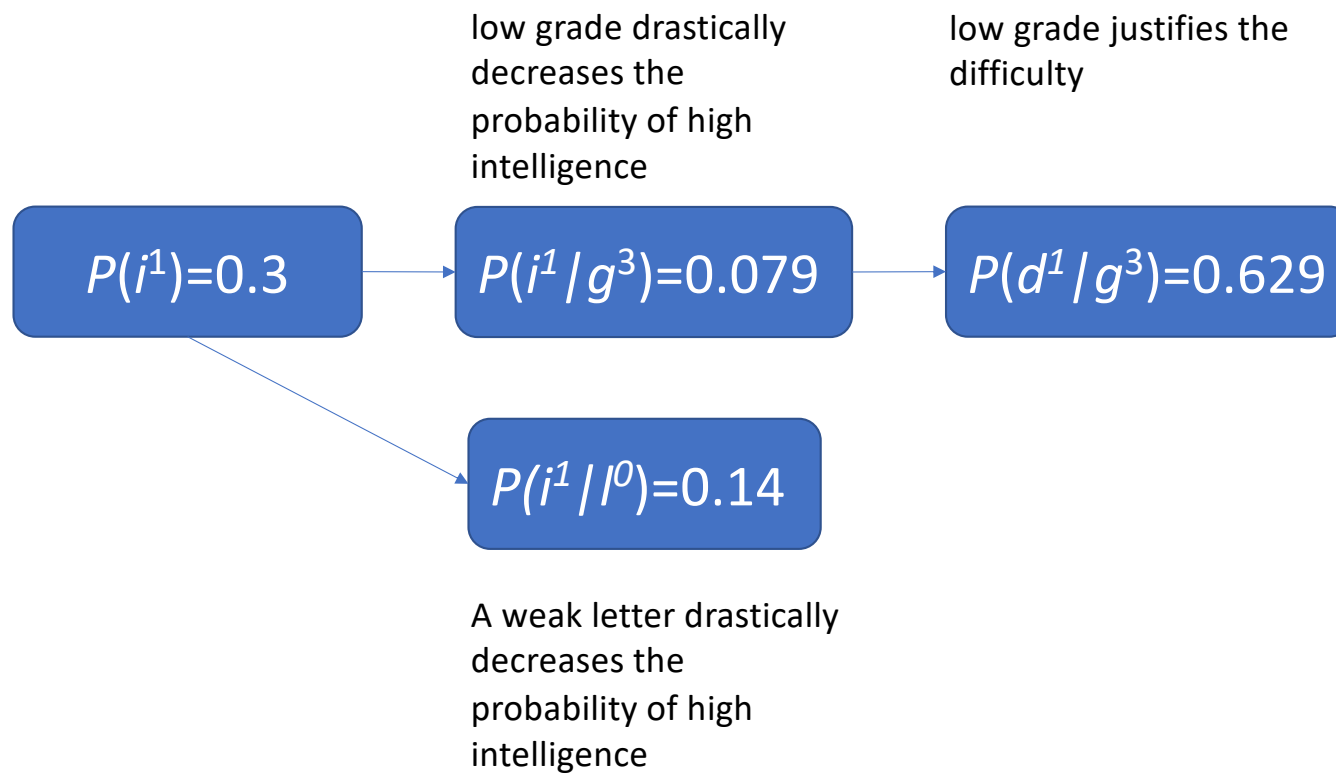
$$\begin{aligned} P(i^1|g^3) &= \frac{P(i^1, g^3)}{P(g^3)} \\ &= \frac{\sum_{D,S,L} P(D)P(i^1)P(g^3|D, i^1)P(S|i^1)P(L|g^3)}{\sum_{D,I,S,L} P(D)P(I)P(g^3|D, I)P(S|I)P(L|g^3)} \end{aligned}$$



Evidential Reasoning

- Recruiter wants to hire *Intelligent* student
- A priori *George* is 30% likely to be *Intelligent*
 $P(i^1)=0.3$
- Finds that *George* received *Grade C* (g^3) in *COMP219*
 $P(i^1 / g^3)=0.079$
- Similarly probability of *Difficult* goes up from 0.4 to
 $P(d^1 / g^3)=0.629$
- If recruiter has lost *Grade* but has *Letter*
 $P(i^1 / l^0)=0.14$



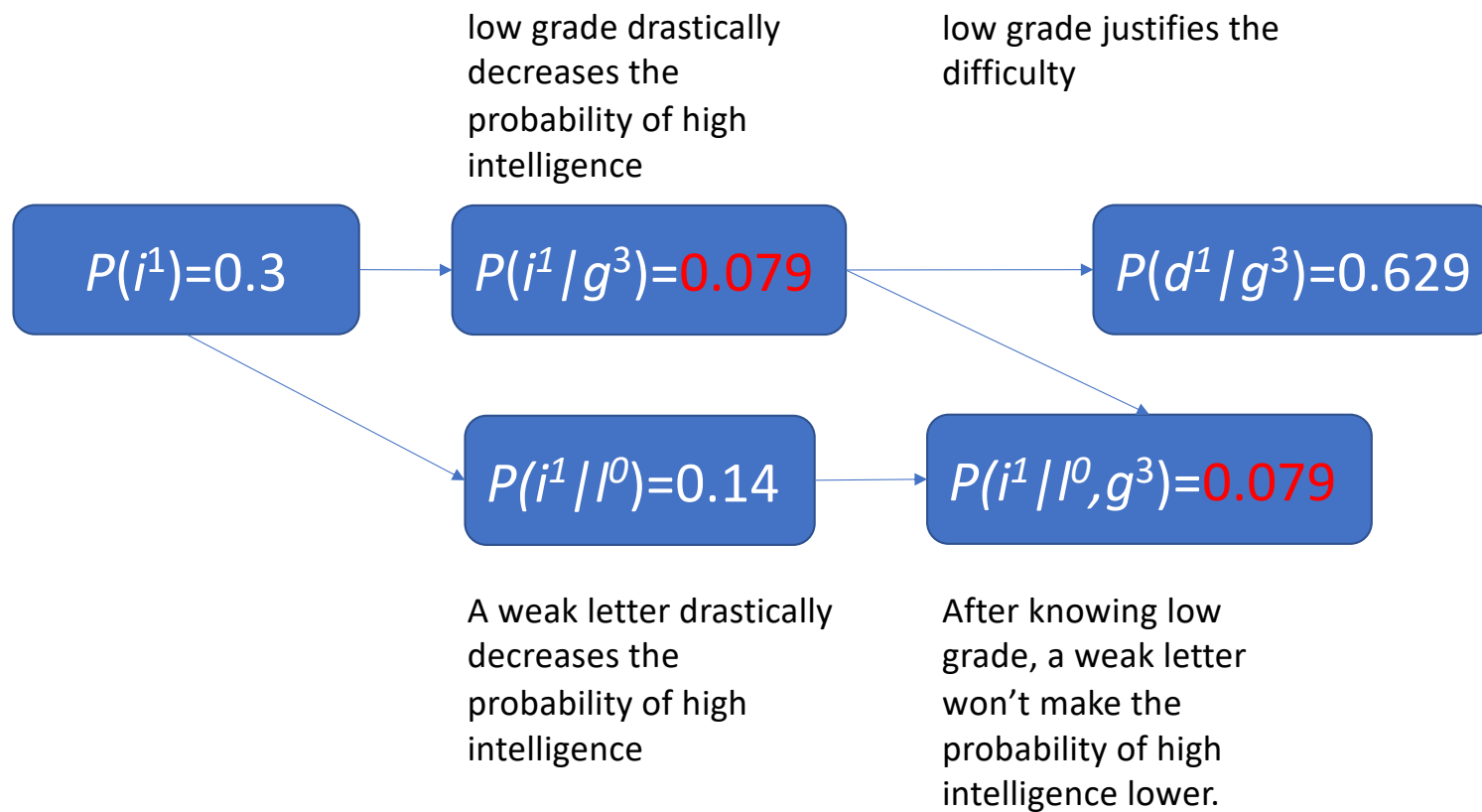


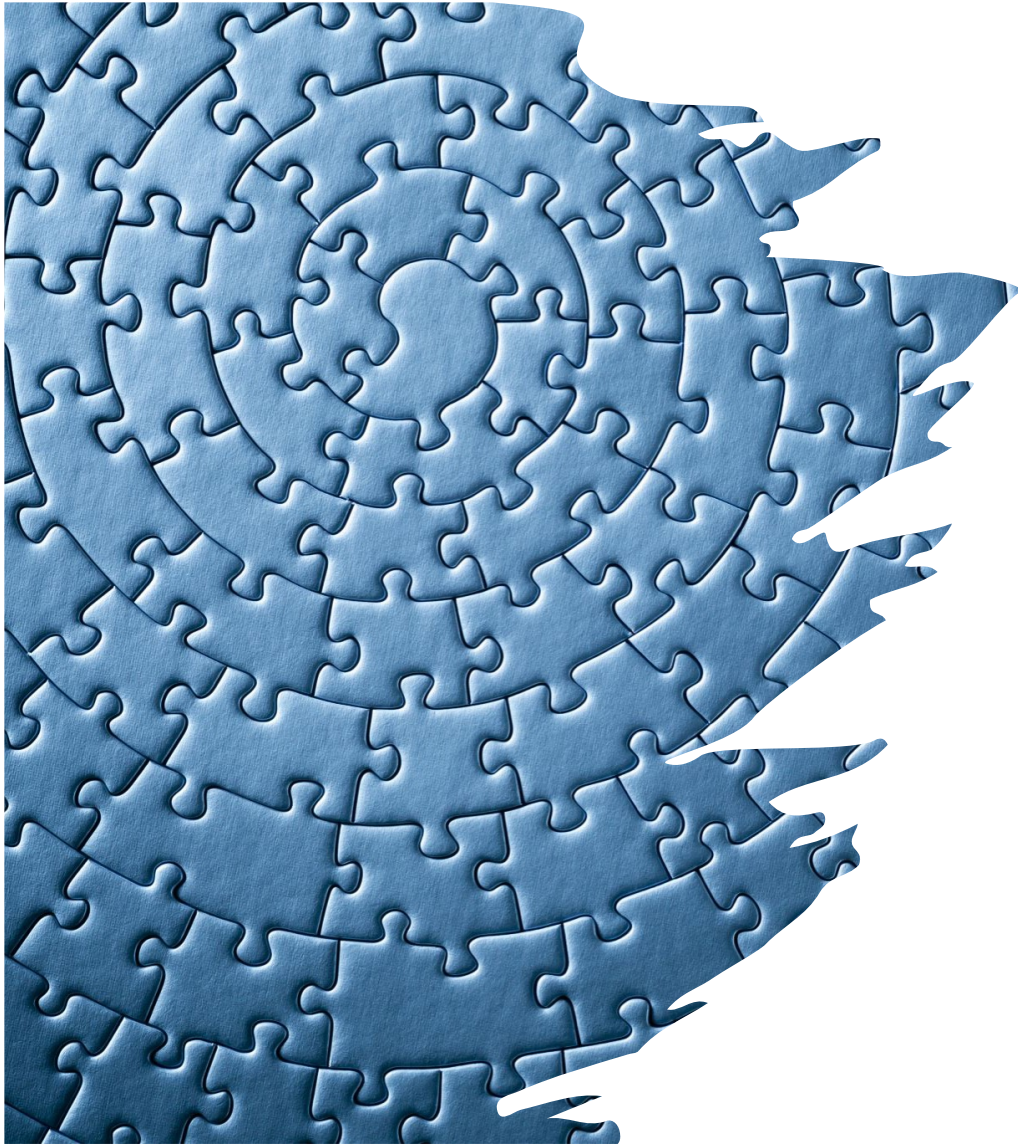
Evidential Reasoning

- Recruiter has both *Grade* and *Letter*

$$P(i^1 | I^0, g^3) = 0.079$$

- Same as if he had only *Grade*
 - *Letter* is immaterial
- Reasoning from effects to causes is called evidential reasoning





Intercausal reasoning

Intercausal reasoning

- Recruiter has Grade (*Letter* does not matter for *Intelligence*)

$$P(i^1/g^3)=P(i^1/l^0,g^3)=0.079$$

- Recruiter receives high *Score* (leads to dramatic increase)

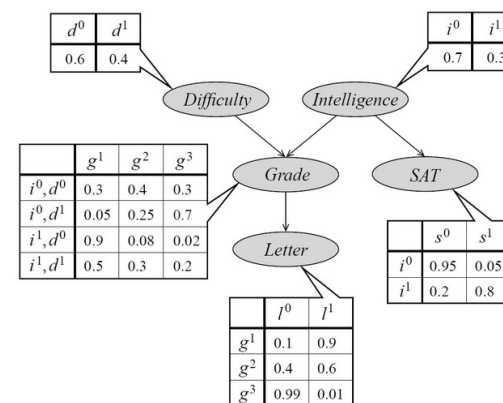
$$P(i^1/g^3,s^1)=0.578$$

- Intuition:

- High Score outweighs poor grade since low intelligence rarely gets good Scores
- Smart students more likely to get Cs in hard classes

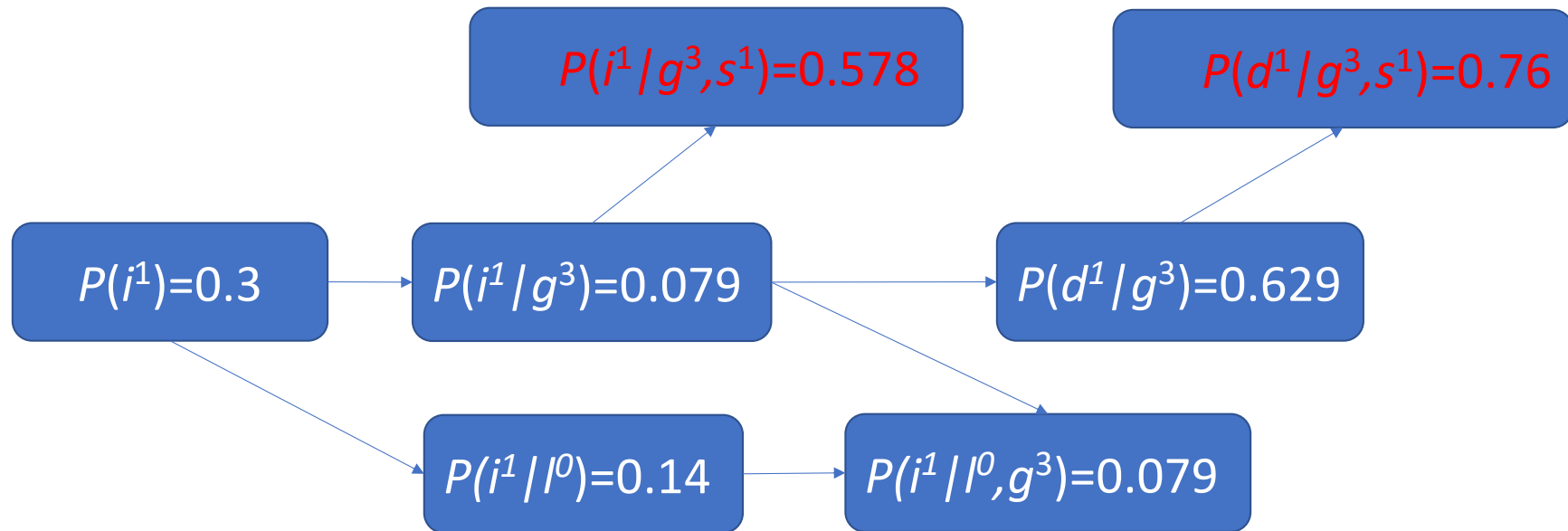
- At the meantime, Probability of class is difficult also goes up from

- $P(d^1/g^3)=0.629$ to
- $P(d^1/g^3,s^1)=0.76$



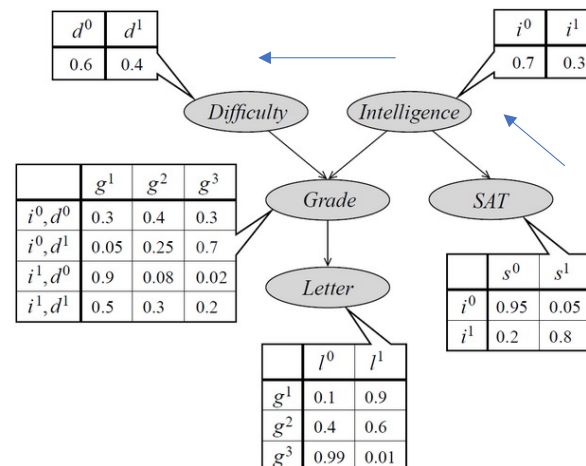
High Score outweighs poor grade since low intelligence rarely gets good Scores

Probability of class is difficult also goes up



Intercausal reasoning

- The previous example:
 - Information about Score gave us information about Intelligence which with Grade told us about difficulty of course
 - One causal factor for **Grade**, i.e., **Intelligence**, give us information about another (**Difficulty**)



Explaining Away

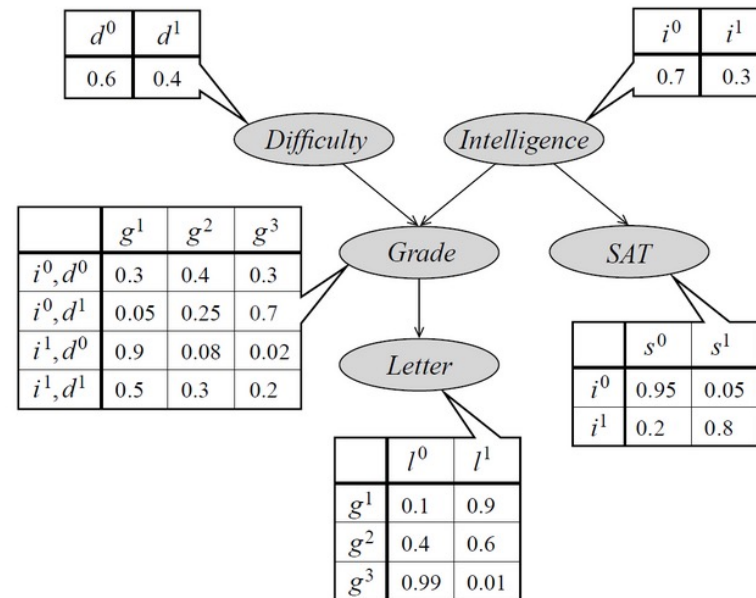
- Given $Grade=C$, $Letter=weak$

$$P(i^1 / g^3) = 0.079$$

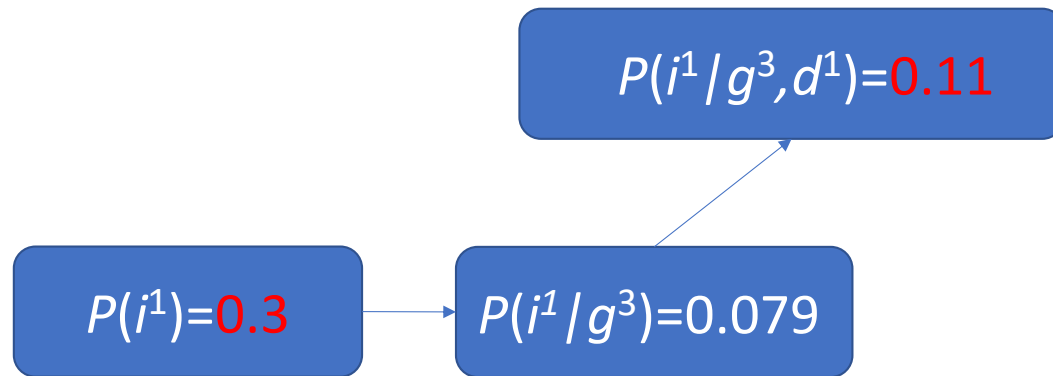
- If we observe $Difficulty=high$

$$P(i^1 / g^3, d^1) = 0.11$$

- We have provided partial explanation for George's grade in COMP219



0.11 < 0.3 : partial explanation
for George's grade



Explaining Away

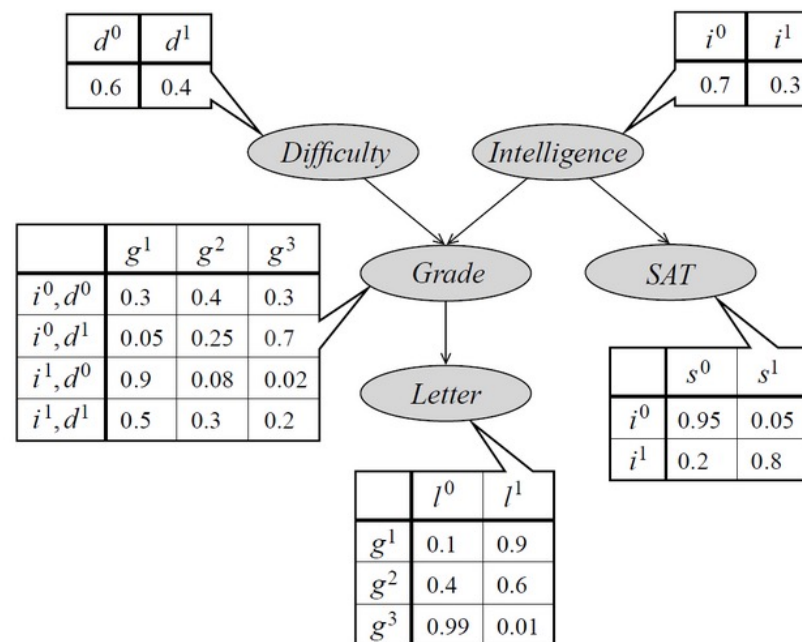
- If George gets a *B* in COMP219

$$P(i^1/g^2)=0.175$$

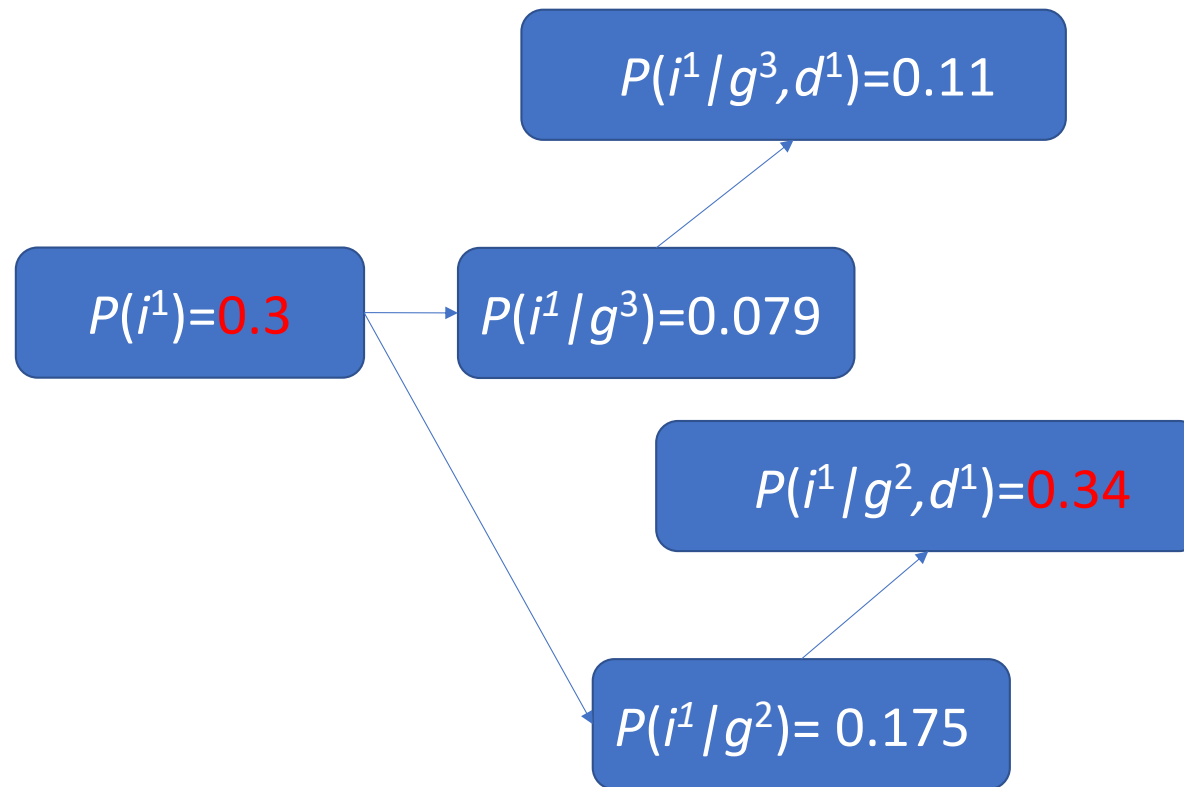
- If we observe COMP219 is hard

$$P(i^1/g^2,d^1)=0.34$$

- We have **explained away the poor grade** via the difficulty of the class



partial explanation for George's
grade



$0.34 > 0.3$:
explained away the
poor grade via the
difficulty of the class

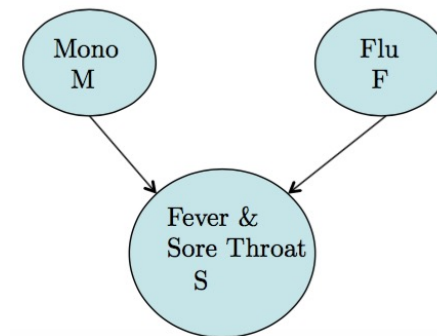
Explaining Away

- Explaining away is one type of intercausal reasoning
- Different causes of the same effect can interact
- All determined by probability calculation rather than heuristics

Common in Human Reasoning

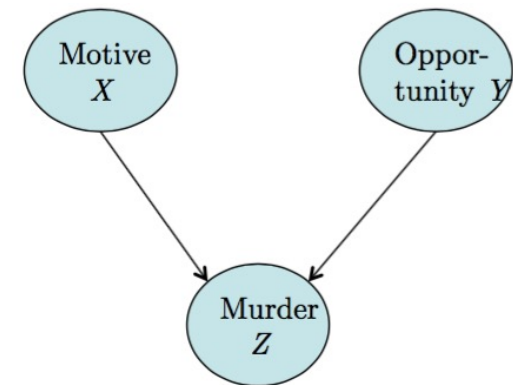
- Binary Variables
- Fever & Sore Throat can be caused by mono and flu
- When flu is diagnosed probability of mono is reduced (although mono could still be present)
- It provides an alternative explanation of symptoms

$$P(m^1/s^1) > P(m^1/s^1, f^1)$$



Another Type of Intercausal Reasoning

- Binary Variables
 - Murder (leaf node)
 - Motive and Opportunity are causal nodes
- Binary Variables X, Y, Z
- X and Y both increase the probability of Murder
 - $P(z^1/x^1) > P(z^1)$
 - $P(z^1/y^1) > P(z^1)$
- Each of X and Y increase probability of the other
 - $P(x^1/z^1) < P(x^1/y^1, z^1)$
 - $P(y^1/z^1) < P(y^1/x^1, z^1)$



Can go in any direction
Different from Explaining
Away